CIE AS PHYSICS (9702) • QUESTION PAPER

4.1 Turning Effects of Forces

Syllabus Section: 4. Forces, Density & Pressure **Paper Type:** Multiple Choice (Paper 1) **Total:** 15 Questions (15 Marks)

Question 1 Easy [1 mark]

What is the centre of gravity of an object?

A the geometrical centre of the object

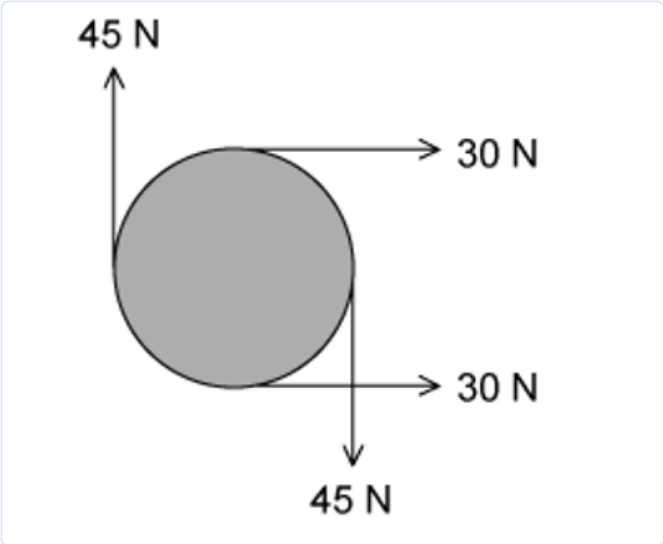
B the point about which the total torque is zero

C the point at which the weight of the object may be considered to act

D the point through which gravity acts

Question 2 Easy [1 mark]

The diagram shows four forces applied to a circular object.



Which of the following describes the resultant force and resultant torque on the object?

	Resultant force	Resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero

A

B

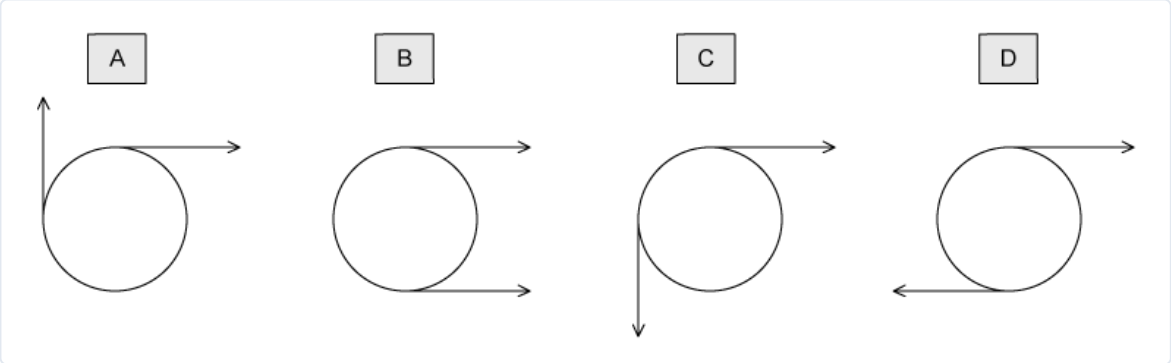
C

D

Question 3 Easy [1 mark]

Two co-planar forces act on the rim of a wheel. The forces are equal in magnitude.

Which arrangement of forces provides only a couple?



A

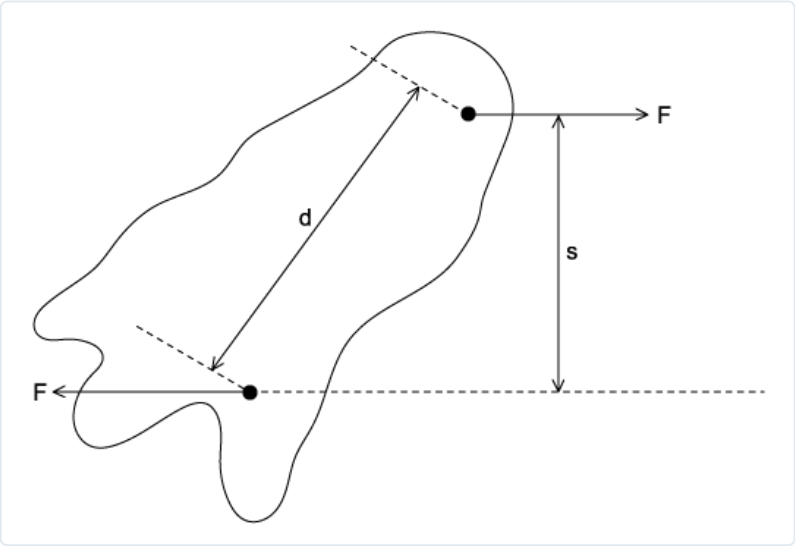
B

C

D

Question 4 Easy [1 mark]

Two parallel forces, each of magnitude *F*, act on a body as shown



What is the magnitude of the torque on the body produced by this couple?

A *Fd*

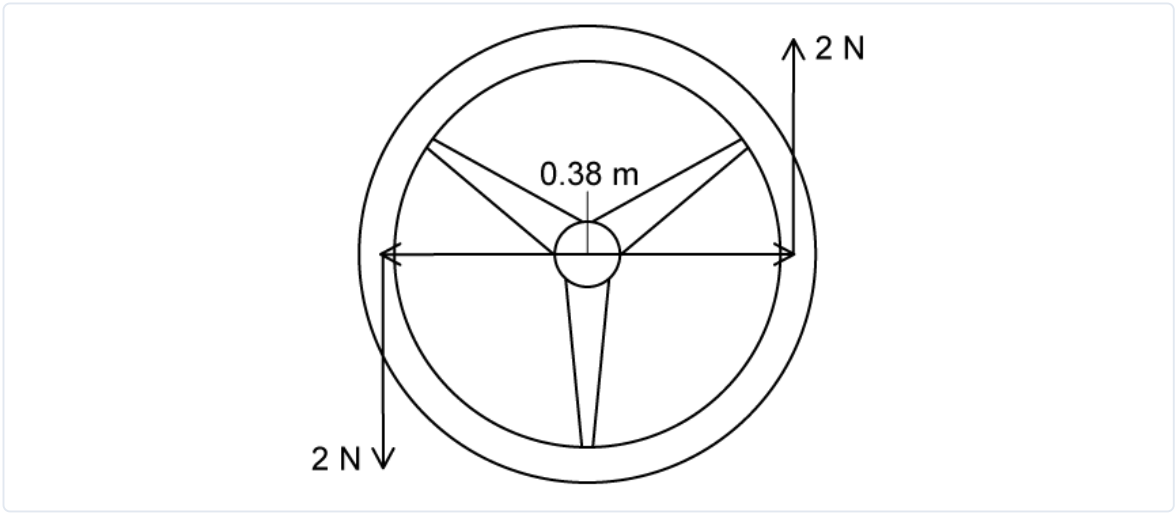
B *Fs*

C *2 Fd*

D *2Fs*

Question 5 Easy [1 mark]

A steering wheel of diameter 0.38 m is rotated to the left by two forces of 2 N, as shown in the diagram.



What is the torque of the couple produced by the rotation of the steering wheel?

A 0.38 N m

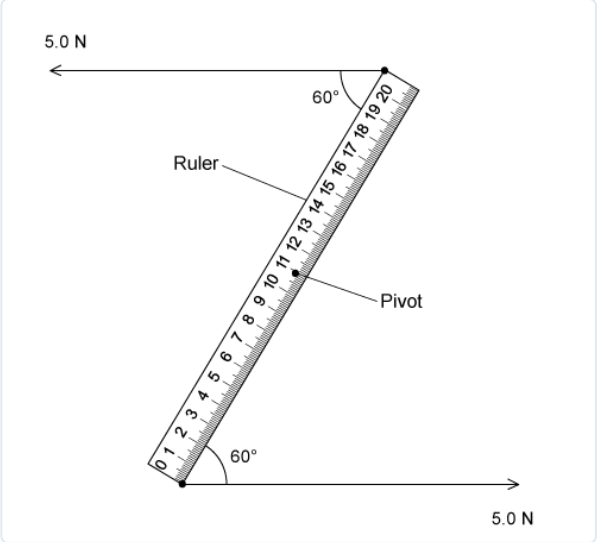
B 0.50 N m

C 0.76 N m

D 1.0 N m

Question 6 Medium [1 mark]

A 20 cm ruler is pivoted at its centre. Equal and opposite forces of magnitude 5.0 N are applied to the ends of the ruler, creating a couple as shown



What is the magnitude of the torque of the couple on the ruler when it is in the position shown?

A 0.2 N m

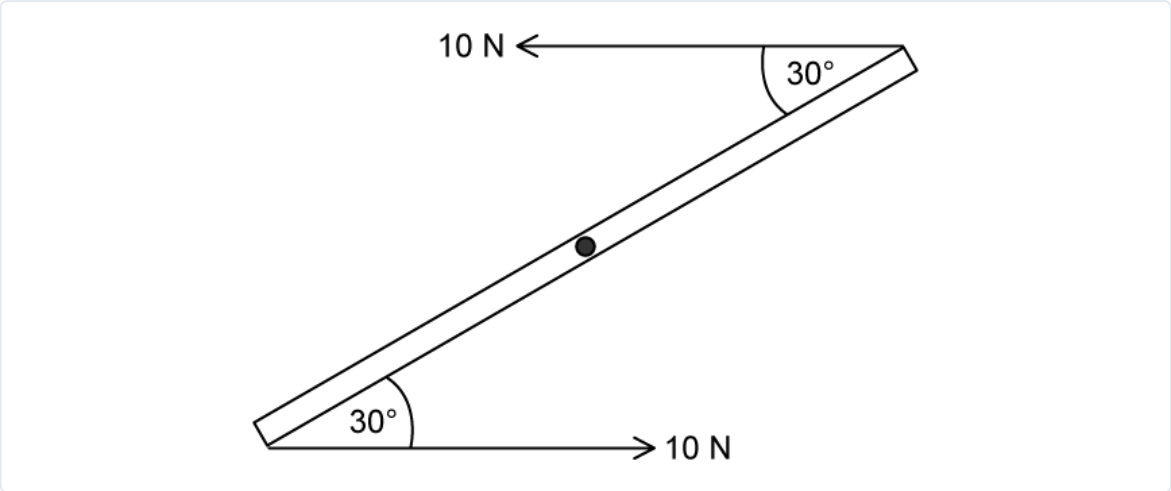
B 0.9 N m

C 1.0 N m

D 1.7 N m

Question 7 Medium [1 mark]

A 10 cm beam is pivoted at its centre. Equal and opposite forces of 10 N pull the ends of the beam, creating a couple as shown.



What magnitude is the torque of the couple?

A 0.3 N m

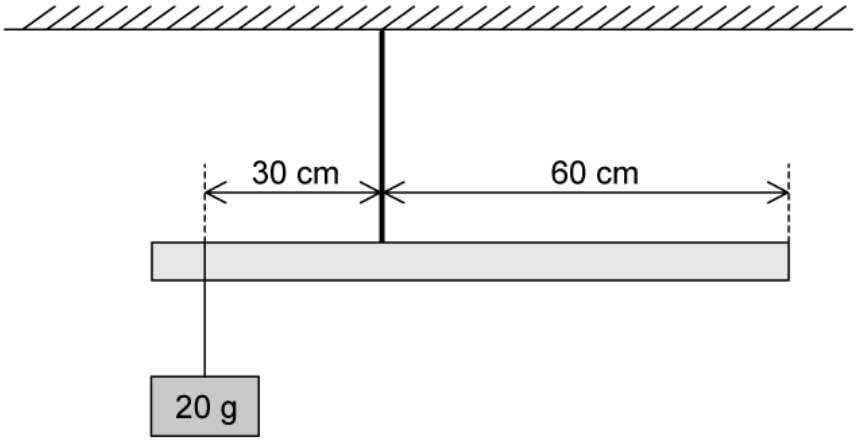
B 0.5 N m

C 0.9 N m

D 1.0 N m

Question 8 Medium [1 mark]

A 1 m long uniform beam is hung from the ceiling by a rope. and is balanced when a 20 g mass is positioned 30 cm from the rope as shown.



What is the mass of the metre rule?

A 0.06 g

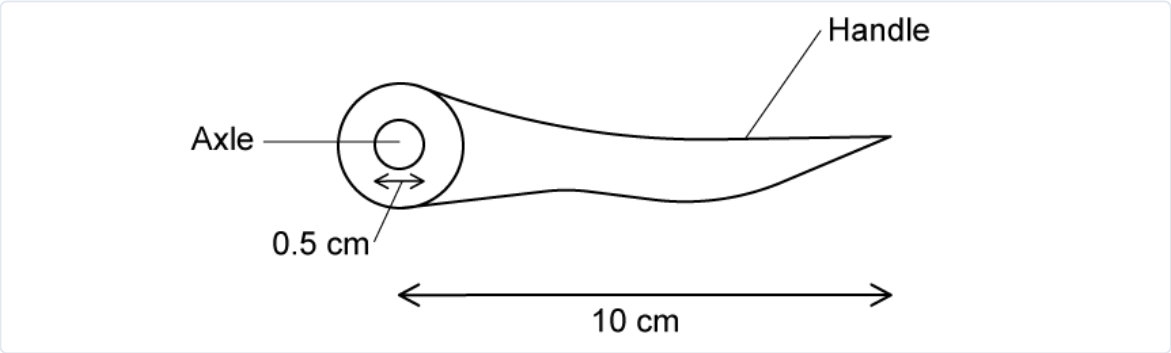
B 20 g

C 60 g

D 100 g

Question 9 Medium [1 mark]

A door handle in the diagram below has a length 10 cm requires a force of 20 N to turn it.



If the handle breaks and the door must be opened using the axle of the handle, with a diameter of 0.5 cm, how much force is needed?

A 4 N

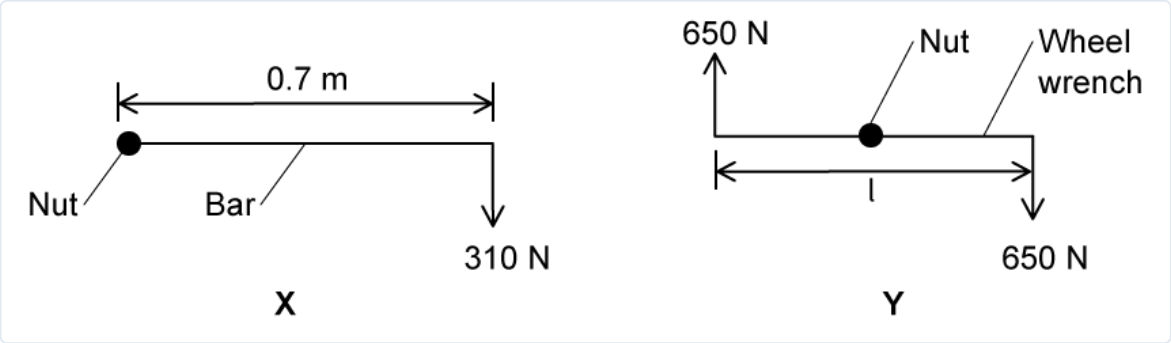
B 8 N

C 40 N

D 400 N

Question 10 Medium [1 mark]

A car wheel nut can be loosened by applying a force of 310 N to the end of a bar of length 0.7 m, as in **X**. A car mechanic is capable of applying forces of 650 N simultaneously in opposite directions on the ends of a wheel wrench, as in **Y**.



What is the minimum length l of the wrench needed for the mechanic to loosen the nut?

A 0.2 m

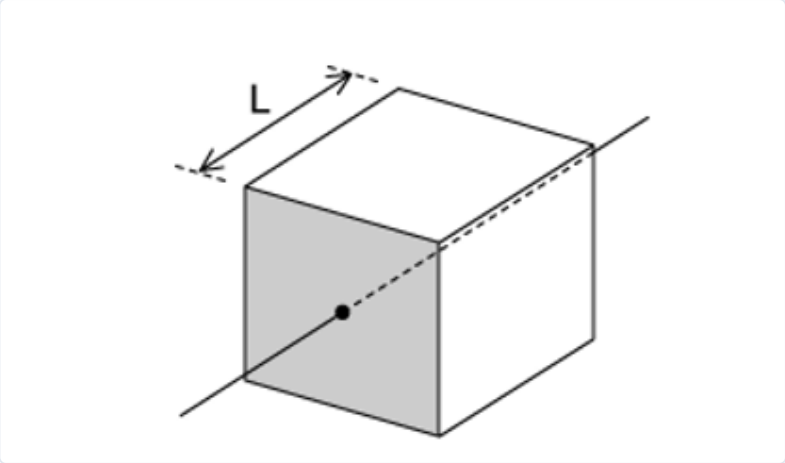
B 0.3 m

C 3.0 m

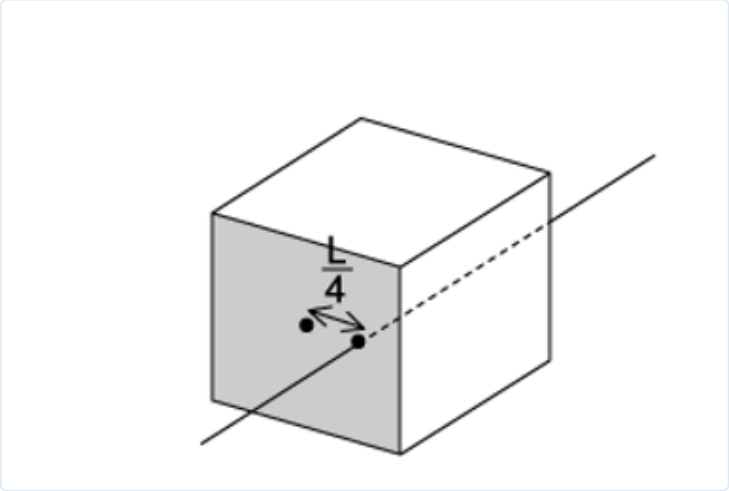
D 6.0 m

Question 11 Hard [1 mark]

The diagram shows a solid cube with weight *W* and sides of length *L*. It is supported by a frictionless spindle that passes through the centres of two opposite vertical faces. One of these faces is shaded.



The spindle is now removed and replaced at a distance $\frac{L}{4}$ to the right of its original position.



When viewing the shaded face, what is the magnitude of the torque required to stop the cube from toppling?

A $\frac{WL}{2}$ anticlockwise

B $\frac{WL}{2}$ clockwise

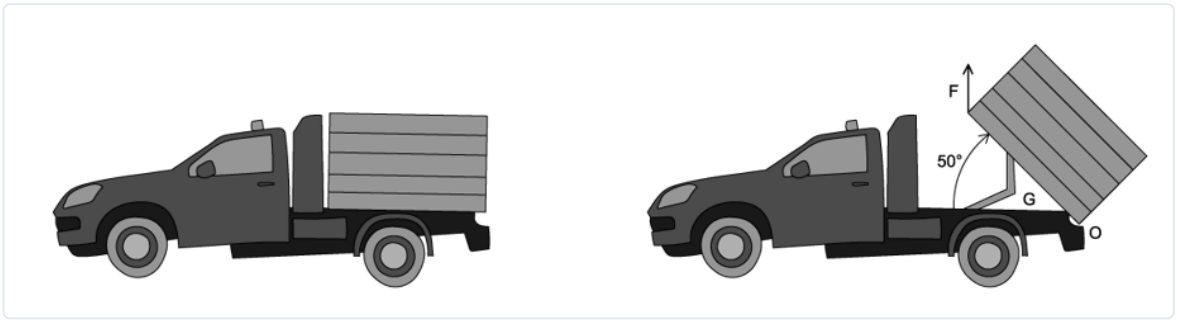
C $\frac{WL}{4}$ anticlockwise

D $\frac{WL}{4}$ clockwise

Question 12 Hard [1 mark]

A trailer of weight 30 kN is attached to a truck. The trailer is raised until it is held at 50˚ to the horizontal by a vertical force *F* as shown in the diagram.

The trailer is 2.7 m in length, and the position of its centre of gravity G is a third of the length from the hinge at O.



What is the vertical force *F* applied to the end of the trailer?

A 10 kN

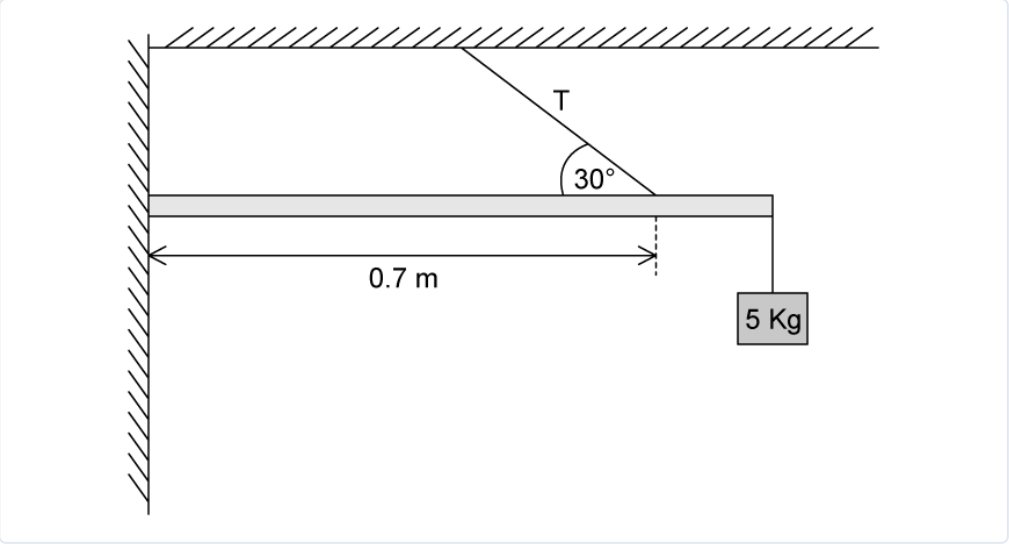
B 15 kN

C 30 kN

D 60 kN

Question 13 Hard [1 mark]

A mass of 5 kg is attached to the end of a 1 m long uniform beam of mass 2 kg. The beam is fixed to a wall at one end and supported by a wire at an angle of 30° to the beam, 0.7 m from the wall. The beam is in equilibrium.



What is the tension *T* in the wire?

A 17 N

B 84 N

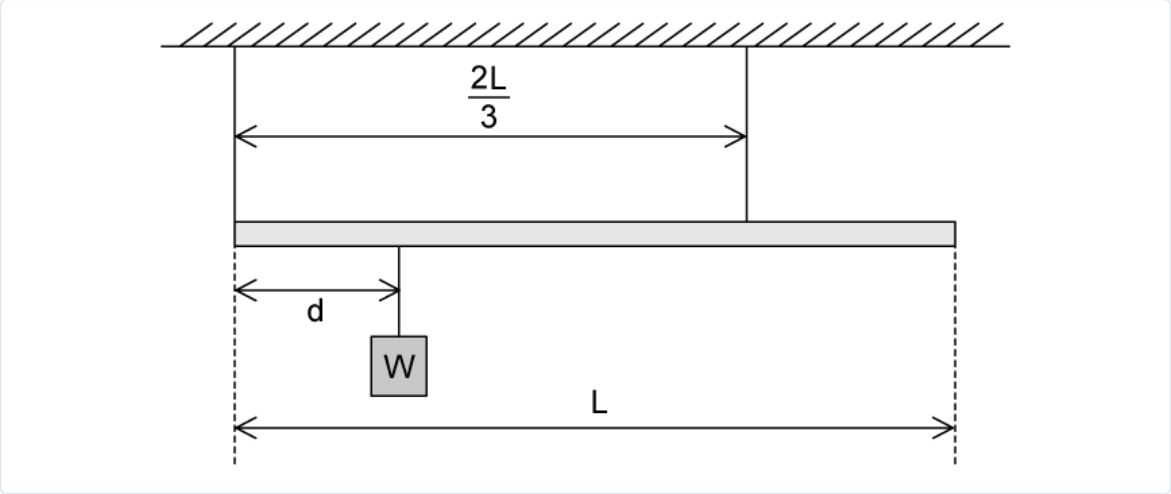
C 140 N

D 170 N

Question 14 Hard [1 mark]

A horizontal beam of length L and weight $\frac{W}{2}$ is suspended from the ceiling by two ropes.

One rope is attached at the left end of the beam, and the other is attached at a distance $\frac{2L}{3}$ from the left end. A load of weight W is suspended from the beam at a distance d from the left end, as shown. The beam is in equilibrium.



What value of d ensures that the tension is equal in both ropes?

A $\frac{L}{8}$

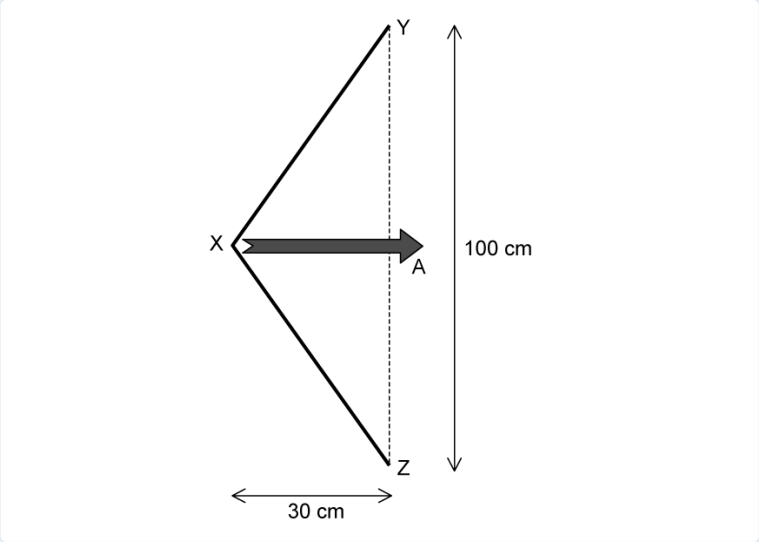
B $\frac{L}{4}$

C $\frac{L}{3}$

D $\frac{L}{2}$

Question 15 Hard [1 mark]

The diagram shows a bow and arrow. The arrow is positioned at the centre X of the string of the bow stretched between points Y and Z. When the string is pulled back, the tension in the bowstring is 10 N.



What force will accelerate the arrow in the direction of point A when the string is released?

A 5.2 N

B 5.8 N

C 8.6 N

D 10.3 N